Assignment 02: Students II Program Report

I. Project Introduction

This assignment implements a CLI that reads course scores for an arbitrary number of students and outputs a summary sheet of student scores and course statistics.

The source code is code/Student2.cpp, and the test data is code/Student2_test.txt.

The runtime environment is Linux 6.6.87.2-microsoft-standard-WSL2, the compiler is g++ 13.3.0, and the program requires C++11 or a newer standard.

II. Running the Program and Input Format

Run the following commands from the assignment's main directory to compile the program and read the test data:

```
g++ -std=c++11 -Wall -Wextra -Wpedantic code/Student2.cpp -o Student2
./Student2 < code/Student2_test.txt
```

Each student record contains a name, a course count, and the corresponding number of “course name, score” pairs. Input continues until the end of file (EOF); when entering data in the terminal, press Ctrl+D to finish. Note: a student may take no courses, but the same student cannot take the same course more than once.

III. Program Output

The test data contains 12 students and 6 courses. The terminal screenshot below shows the compilation command, execution command, and actual output described above:



```
(base) yifeng@maple:~/projects/OOP/Assignment02_3250105856_ZhengYifeng$ g++ -std=c++11 -Wall -Wextra -Wpedantic code/Student2.cpp -o Student2
(base) yifeng@maple:~/projects/OOP/Assignment02_3250105856_ZhengYifeng$ ./Student2 < code/Student2_test.txt
no      name    Math    English Physics Programming     Art     History average
1       Alice   5       4       3       -       -       -       4.00
2       Bob     3       -       -       5       -       -       4.00
3       Carol   -       2       -       -       -       -       2.00
4       David   4       5       1       4       -       -       3.50
5       Eve     -       -       -       -       -       -       -
6       Frank   -       -       5       3       -       -       4.00
7       Grace   1       -       -       -       5       -       3.00
8       Henry   5       -       -       -       -       -       5.00
9       Ivy     -       1       -       -       1       5       2.33
10      Jack    -       -       -       1       -       1       1.00
11      Kate    2       3       4       2       3       4       3.00
12      Leo     -       -       -       -       -       -       -
        average 3.33    3.00    3.25    3.00    3.00    3.33
        min     1       1       1       1       1       1
        max     5       5       5       5       5       5
```

The summary sheet displays student numbers, names, scores for each course, and individual averages in that order. The bottom rows display the average, minimum, and maximum score for each course.

If a student has not taken a course, - is displayed; if a student takes zero courses, the average is displayed as -.

IV. Conclusion

The program uses the Student and Course classes to handle student data and course statistics respectively, outputs a summary sheet similar to Students I, and completes the assignment.